

Lesson 6

Configuring Storage Management

Topic 6A

Manage Storage

Exam Objectives Covered

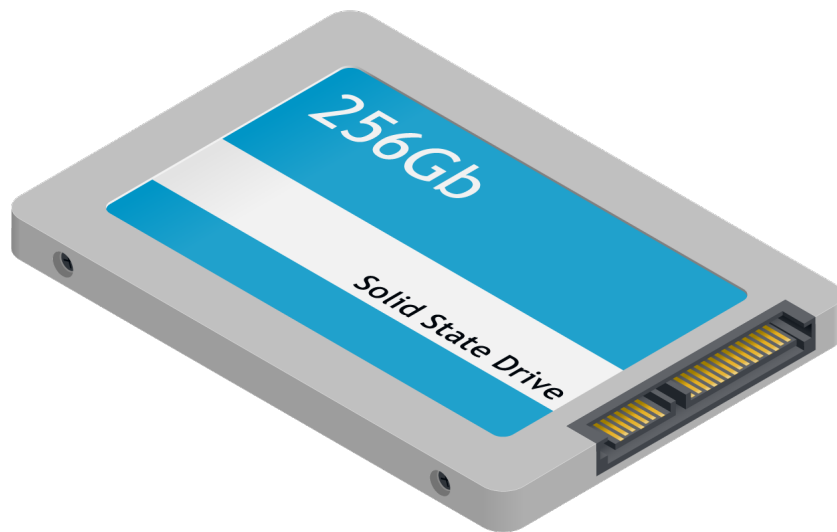
- 1.2 Given a scenario, deploy and manage storage
- 2.3 Given a scenario, configure and maintain server functions and features

Capacity Planning

- Considerations
- Enforce capacity management
 - Quotas
 - Deduplication
 - Compression
- Storage capacity and the cloud

Hard Drive Media Types

- HDDs
- SSDs
- Hybrid
 - Dual-drive
 - SSD
- SSD versus HDD



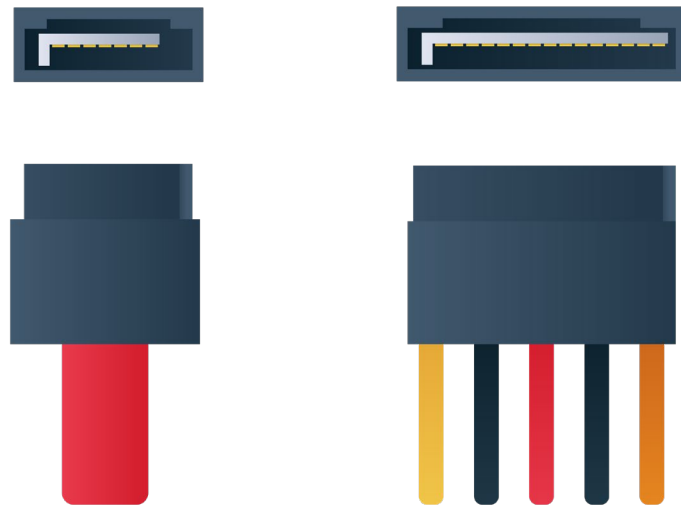
Solid-state drive. (Image © 123RF.com)

HDD Interface Types

- Direct Attached Storage (DAS)
 - SATA
 - SAS
 - NVMe

DAS Controllers and Drives

- SATA



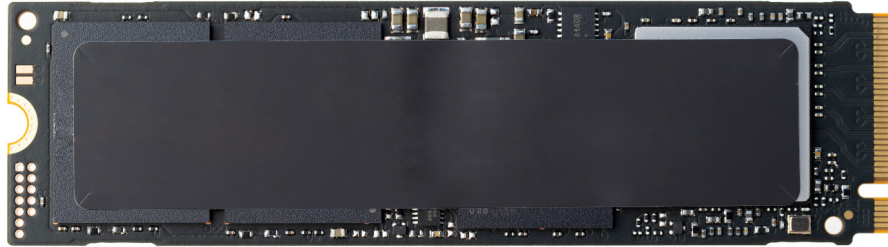
SATA cables. SATA data cables are typically smaller and usually contain 7 pins, whereas SATA power cables are larger and contain 15 pins. (Image © 123RF.com)

DAS Controllers and Drives (slide 2)

- SAS
- NVMe

DAS Drive Connectors

- M.2 specification



SSD M.2 form factor. (Image by Vitaly Pozdeyev © 123RF.com)

DAS Drive Connectors (2)

- PCIe
- eSATA
- USB

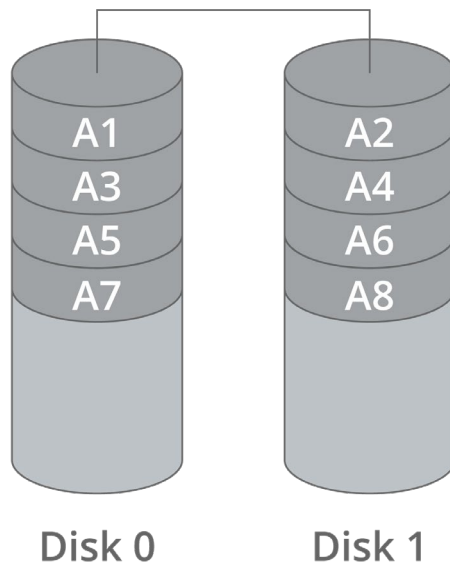


USB Type C connector. (Image by tevarak11 © 123RF.com)

- SD cards

RAID Levels and Types

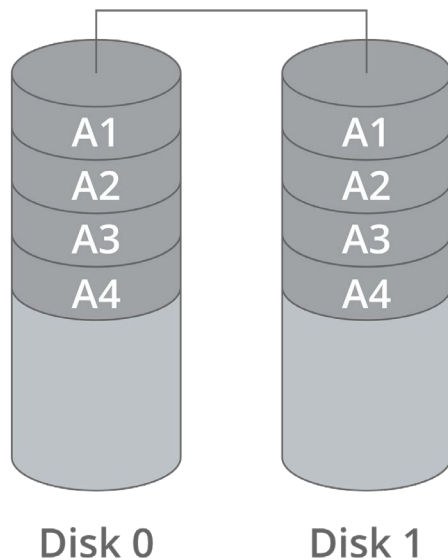
- RAID 0 disk striping



RAID 0 disk striping. (Images © 123RF.com)

RAID Levels and Types (slide 2)

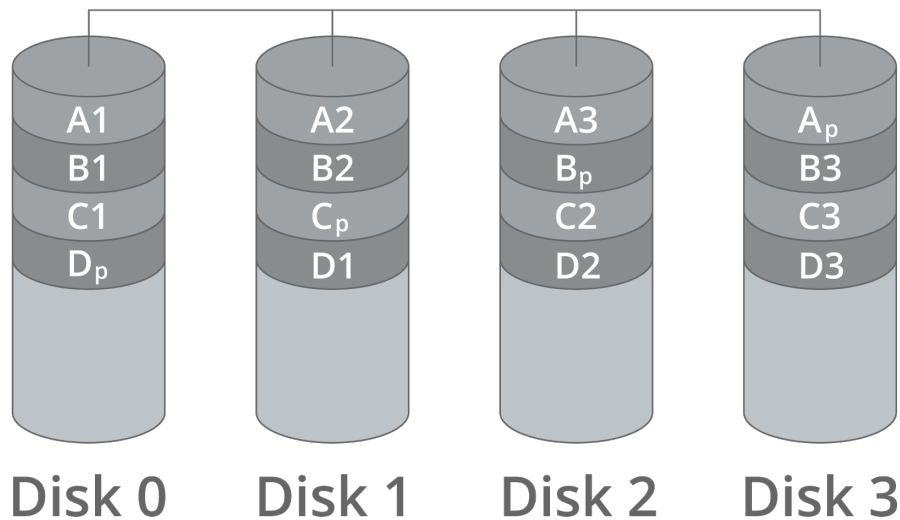
- RAID 1 disk mirroring



RAID 1 disk mirroring. (Images © 123RF.com)

RAID Levels and Types (slide 3)

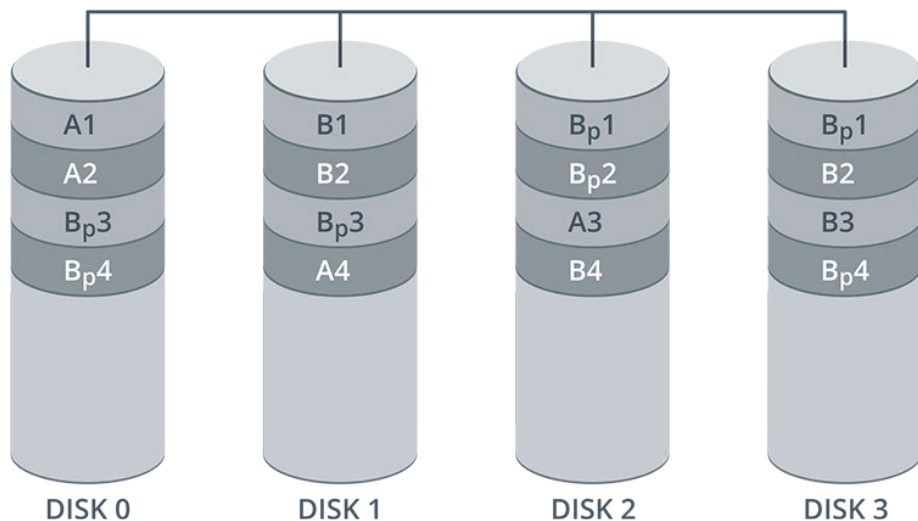
- RAID 5 disk striping with parity



RAID 5 disk striping with parity. (Images © 123RF.com)

RAID Levels and Types (slide 4)

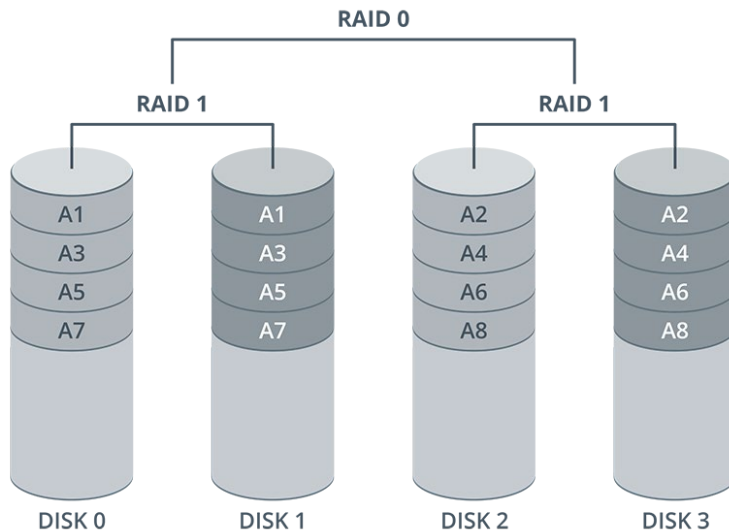
- RAID 6 disk striping with double parity



RAID 6 disk striping with double parity. (Images © 123RF.com)

RAID Levels and Types (slide 5)

- RAID 10 disk mirroring with striping



RAID 10 disk mirroring with striping. (Images © 123RF.com)

RAID Levels and Types (6)

- JBOD
- Software RAID vs. Hardware RAID

Shared Storage

- DAS
- NAS
- SAN

Network File Sharing Protocols

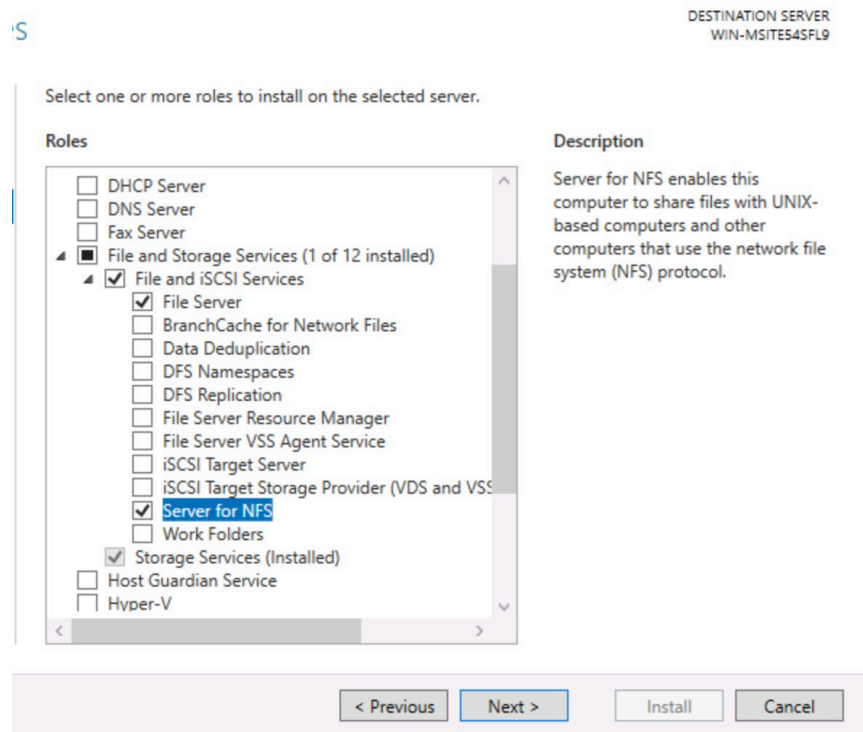
- NFS

```
# Export the /projects directory with rw access to workstation01  
/projects          workstation01 (rw,sync)
```

Exporting the /projects directory for network access.

Network File Sharing Protocols (slide 2)

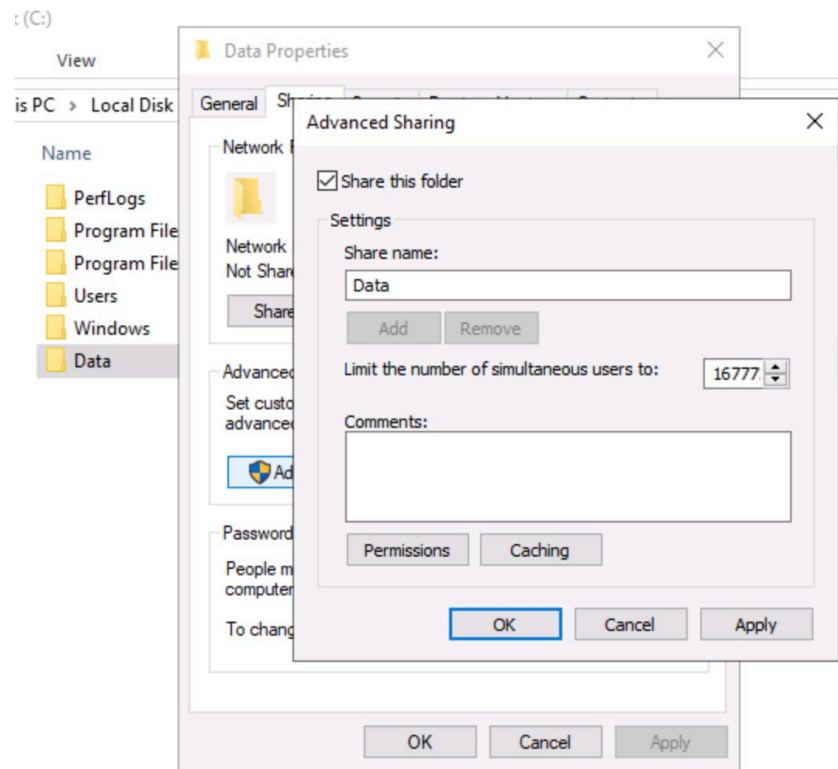
- NFS client for Windows



Add the Server for NFS Role to Windows Server 2019. (Screenshot courtesy of Microsoft.)

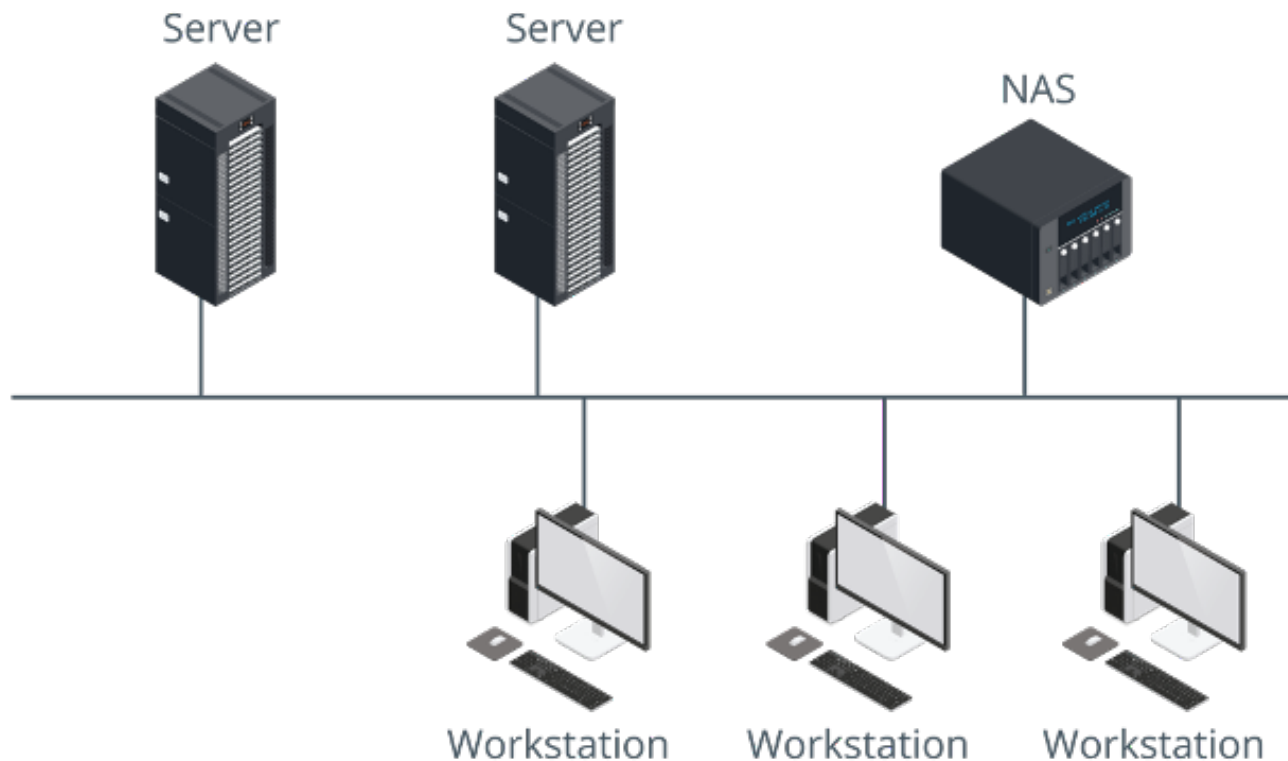
Network File Sharing Protocols (slide 3)

- SMBs



Sharing a folder in Windows. (Screenshot courtesy of Microsoft.)

Network Attached Storage (NAS)



Network attached storage. (Images © 123RF.com)

SAN

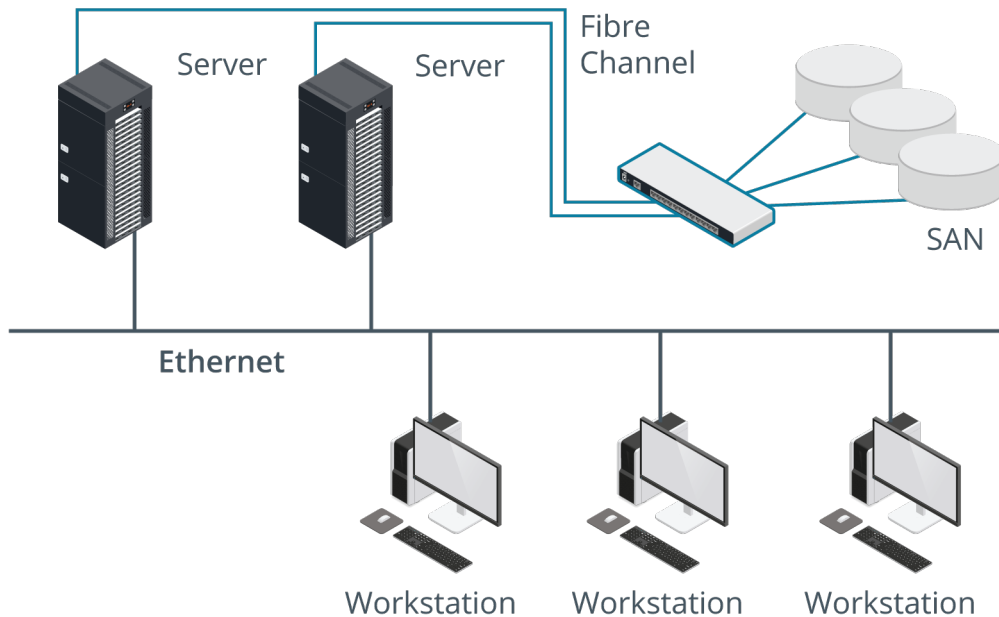
- HBA
- Connector types for SANs



Small form-factor pluggable (SFP) connectors. (Image © 123RF.com)

SAN (slide 2)

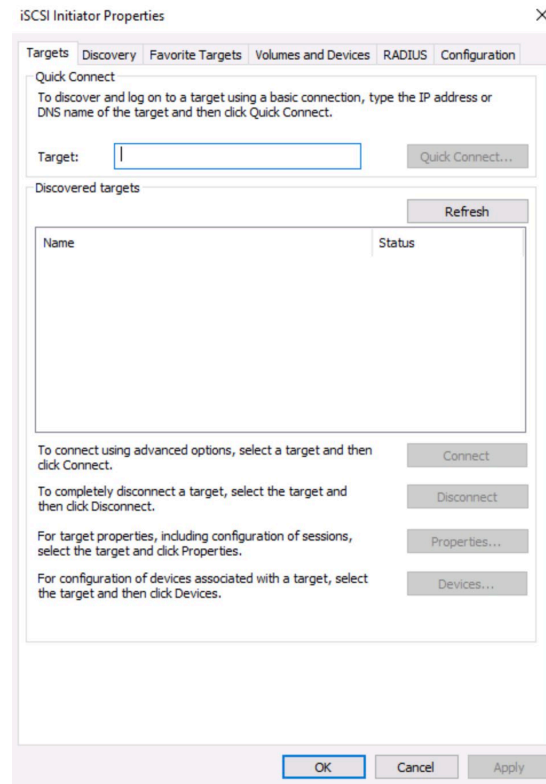
- Protocols
- Fibre channel



Switched SAN. (Image © 123RF.com)

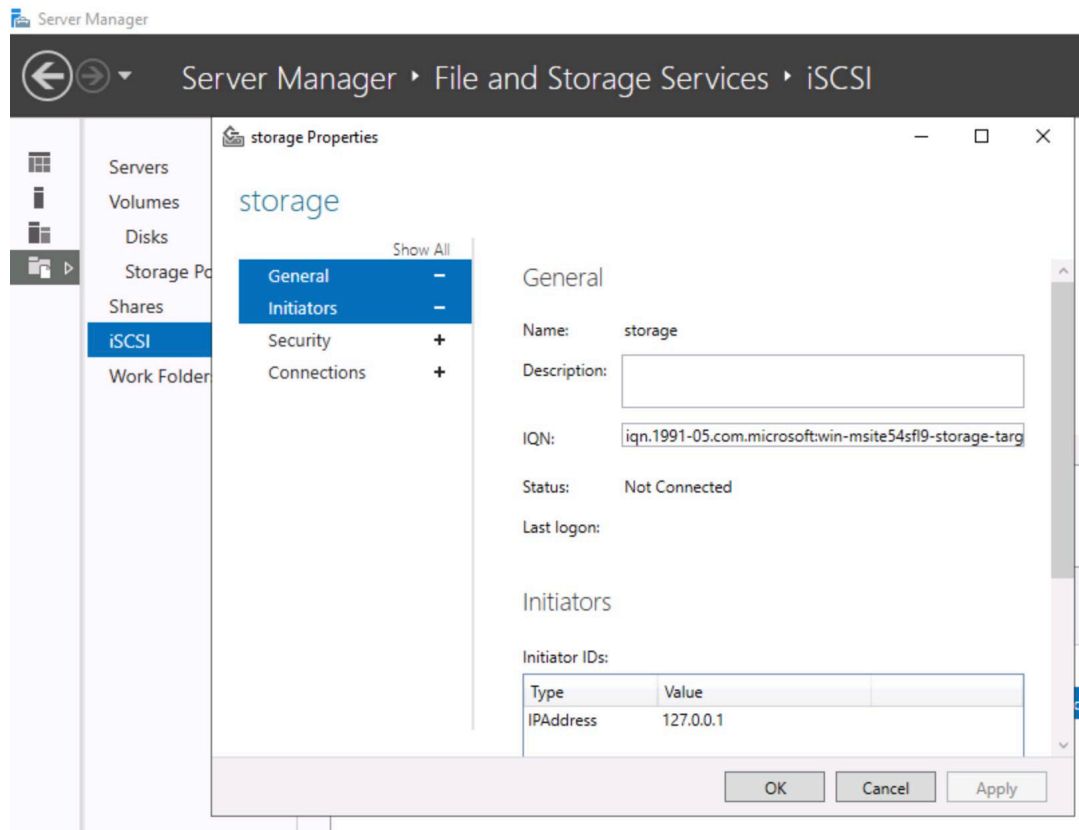
SAN (slide 3)

- Fibre channel over Ethernet (FCoE)
- Internet Small Computers Systems Interface (iSCSI)



*iSCSI Initiator Properties in Windows Server 2019.
(Screenshot courtesy of Microsoft.)*

SAN (slide 4)



Configuring Windows Server as an iSCSI target. (Screenshot courtesy of Microsoft.)

Guidelines for Managing Storage

- RAID 0
- RAID 1
- RAID 5
- DAS
- NFS
- SMB
- SAN
- iSCSI
- Fibre channel

Review Activity: Storage Management



Review Activity

Topic 6B

Troubleshoot Storage

Exam Objectives Covered

- 2.3 Given a scenario, configure and maintain server functions and features
- 4.3 Given a scenario, troubleshoot storage problems

Common Storage Problems

- Physical connectivity
- Drive configuration
- RAID arrays
- Capacity

Tools and Techniques for Troubleshooting Storage

- Partition management

```
Command (m for help): m
Help:

DOS (MBR)
a toggle a bootable flag
b edit nested BSD disklabel
c toggle the dos compatibility flag

Generic
d delete a partition
F list free unpartitioned space
l list known partition types
n add a new partition
p print the partition table
t change a partition type
v verify the partition table
i print information about a partition

Misc
m print this menu
u change display/entry units
x extra functionality (experts only)

Script
I load disk layout from sfdisk script file
O dump disk layout to sfdisk script file

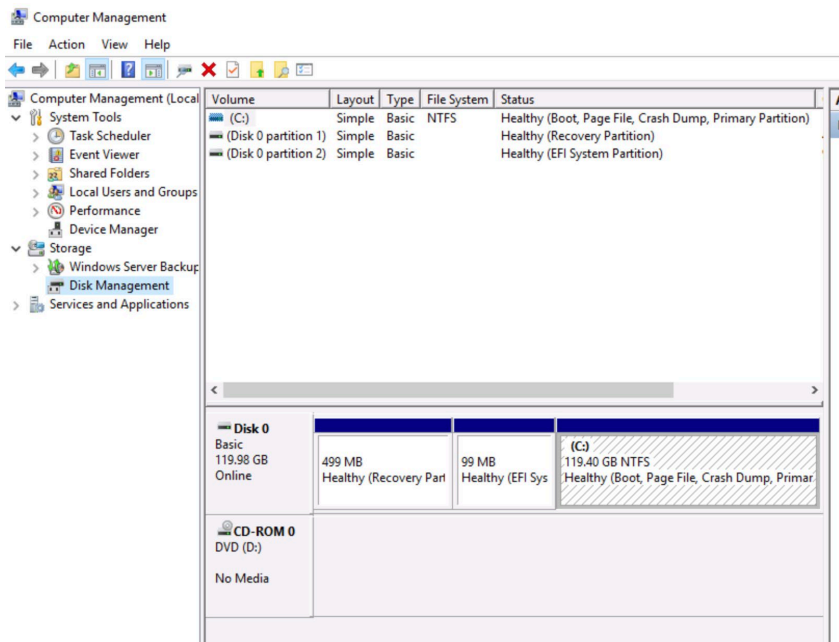
Save & Exit
w write table to disk and exit
q quit without saving changes

Create a new label
g create a new empty GPT partition table
G create a new empty SGI (IRIX) partition table
o create a new empty DOS partition table
s create a new empty Sun partition table
```

fdisk options menu.

Tools and Techniques for Troubleshooting Storage (slide 2)

- Disk management/Drive monitoring



Windows Server Disk Management. (Screenshot courtesy of Microsoft.)

Tools and Techniques for Troubleshooting Storage (slide 3)

- RAID array configuration
- Partition Mounting Commands

```
PS C:\Users\Administrator> net share
```

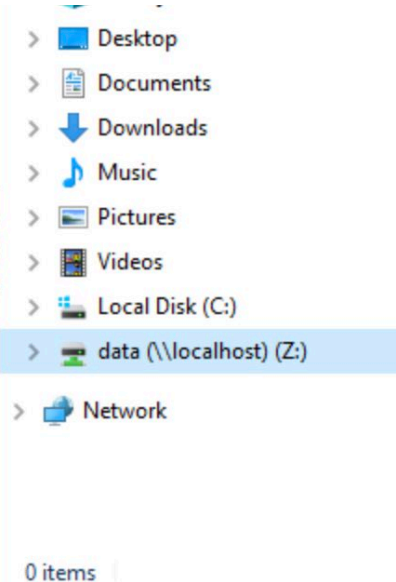
Share name	Resource	Remark
C\$	C:\	Default share
IPC\$		Remote IPC\$
ADMIN\$	C:\Windows	Remote Administrator
Data	C:\Data	

```
The command completed successfully.
```

```
PS C:\Users\Administrator> net use Z: \\localhost\data
```

```
The command completed successfully.
```

```
PS C:\Users\Administrator>
```



The screenshot shows a Windows File Explorer window. The left sidebar displays the 'This PC' view with various folders like Desktop, Documents, Downloads, Music, Pictures, Videos, Local Disk (C:), and Network. The 'data' folder, which is mapped to the Z: drive, is highlighted in blue. The main pane shows '0 items'.

net share and net use commands, with mapped Z: drive. (Screenshot courtesy of Microsoft.)

Tools and Techniques for Troubleshooting Storage (slide 4)

- System logs

```
[root@localhost ~]# ls /var/log
anaconda      chrony      firewallld  journal     README      tallylog      wtmp
audit         cups        gdm         lastlog     samba       vboxadd-install.log
blivet-gui    dnf.librepo.log glusterfs   libvirt     speech-dispatcher
boot.log      dnf.log     hawkey.log  ppp         sssd        vboxadd-setup.log
btmptmp      dnf.rpm.log httpd       private     swtpm       vboxadd-setup.log.1
[root@localhost ~]#
```

- Visual and auditory inspections

The Linux /var/log directory.

Review Activity: Common Storage Problems



Review Activity

Lesson 6

Summary

